Amplifier Class Stage

1. An amplifier that operates in the linear region at all times is

(a) Class A (b) Class AB (c) Class B (d) Class C

Answer A

2. A certain class A power amplifier delivers 5 W to a load with an input signal power of

100 mW. The power gain is

(a) 100 (b) 50 (c) 250 (d) 5

Answer b

5. A certain class A power amplifier has the maximum signal power output is

(a) 6 W (b) 12 W (c) 1 W (d) 0.707 W

Answer C

6. The efficiency of a power amplifier is the ratio of the power delivered to the load to the

(a) input signal power (b) power dissipated in the last stage

(c) power from the dc power supply (d) none of these answers

Answer C

7. The maximum efficiency of a class A power amplifier is

(a) 25% (b) 50% (c) 79% (d) 98%

a

The transistors in a class B amplifier are biased

(a) into cutoff (b) in saturation

(c) at midpoint of the load line (d) right at cutoff

Answer d

9. Crossover distortion is a problem for

(a) class A amplifiers (b) class AB amplifiers

(c) class B amplifiers (d) all of these amplifiers

Answer b

Class A Amplifier: No Crossover Distortion occurs as they are biased in the center of the load line. Class B Amplifiers: Large amounts of Crossover Distortion due to biasing at the cut-off point. Class AB Amplifiers: Crossover Distortion is avoided as the biasing level is above the cut-off.

12. The maximum efficiency of a class B push-pull amplifier is

(a) 25% (b) 50% (c) 79% (d) 98%

Answer C

The maximum efficiency of a class AB amplifier is

(a) higher than a class B (b) the same as a class B

(c) about the same as a class A (d) slightly less than a class B

Answer b

The efficiency of a class C amplifier is

(a) less than class A (b) less than class B

(c) less than class AB (d) greater than classes A, B, or AB

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d

The transistor in a class C amplifier conducts for

(a) more than of the input cycle (b) one-half of the input cycle

(c) a very small percentage of the input cycle (d) all of the input cycle

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C
